

UNIGRAM-CALIBRATED STATIONARY DISCRETE DIFFUSION LANGUAGE MODELS

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ABSTRACT

Isotropic uniform noise corruption in non-masked and mixed discrete diffusion language models (XDLM) forces intermediate sequence states off the natural-language manifold, wasting representation capacity on high-entropy, unnatural token combinations. In this paper, we propose **Unigram-Calibrated Stationary Discrete Diffusion (UNI-XDLM)**, a discrete diffusion framework that replaces white uniform noise with an empirical Zipfian stationary target distribution p_γ . Crucially, we prove that substituting isotropic uniform targets with an empirical unigram prior preserves a *100% exact* closed-form scalar Evidence Lower Bound (ELBO) that is evaluated in $O(|V|)$ time, completely bypassing variational bit-factorizations, matrix multiplications, or Taylor truncations. Evaluated on the OpenWebText pretraining corpus and a comprehensive 7-dataset zero-shot suite under compute-matched conditions, UNI-XDLM attains an in-domain validation perplexity of 146.4 (versus baseline XDLM’s 155.3 and UDLM’s 164.2) and reduces the average zero-shot perplexity to 380.7 (versus 406.4 for MDLM, 432.6 for XDLM, and 463.7 for UDLM)—a +51.8 point (12.0%) gain over XDLM from a single, parameter-free design change. The advantage *widens with scale*: at sequence length 512 the margin over XDLM grows to +20.4 PPL and UNI-XDLM ranks second among seven methods including the SoTA baselines SEDD and RADD, while pure masking collapses. We further derive an ELBO-preserving reverse-rate anneal that removes multi-step diversity collapse without post-hoc truncation, show through a matched-entropy ablation that vocabulary-support preservation—not entropy—drives density, and verify at 8B scale that continual pretraining of LLaDA-8B preserves MMLU and HumanEval. Six controlled ablations calibrate the exponent, schedules, collapse remedy, prior corpus, smoothing, and an inward-bowing understanding-generation Pareto frontier with a sweet spot at $k_0 = 0.1$.

1 INTRODUCTION

Generative modeling of discrete sequences, particularly natural language, has long been dominated by autoregressive (AR) architectures that predict text token-by-token from left to right. Recently, non-autoregressive discrete diffusion language models (DDLMS) have emerged as a promising alternative, offering non-causal generation, parallel sampling, and flexible controllable text synthesis (Austin et al., 2021; Li et al., 2022; Hoogetboom et al., 2021b; Dieleman et al., 2022). To model categorical structures, researchers have explored continuous-space diffusions (Dieleman et al., 2022; Avdeyev et al., 2023), simplex-based probability paths (Cheng et al., 2024; Song et al., 2025), discrete flow matching (Gat et al., 2024), and discrete Markov jump processes in continuous time (Campbell et al., 2022; Hoogetboom et al., 2021a). When scaled to large token budgets and parameter counts, discrete diffusion architectures exhibit scaling behaviors competitive with autoregressive models (Nie et al., 2024; Bie et al., 2025).

Despite these advances, standard mixed and non-masked discrete diffusion frameworks—such as Uniform Discrete Diffusion Models (UDLM) (Lou et al., 2023) and mixed discrete diffusion models (XDLM) (Liu et al., 2026)—rely fundamentally on isotropic *white uniform* noise kernels during sequence corruption. Corrupting categorical states toward an uninformative uniform distribution forces intermediate noisy states z_t into unnatural, high-entropy token configurations that drift severely off the natural-language manifold. To mitigate this degradation, existing approaches

resort to continuous bit-channel factorizations (Chen et al., 2022), score-ratio Taylor approximations (Zhang et al., 2025), or full-sequence energy-based rescaling (Xu et al., 2024). However, factorizing discrete states introduces severe structural total-correlation variational gaps, while continuous approximations degrade density estimation without guaranteeing closed-form likelihoods.

In this paper, we propose **Unigram-Calibrated Stationary Discrete Diffusion (UNI-XDLM)**, a framework that replaces white uniform noise targets with an empirical Zipfian unigram stationary kernel K_{uni} . Inspired by the power-law token distributions of natural corpora (Mikolov et al., 2013), p_γ keeps corrupted tokens z_t grounded on the language manifold by replacing tokens with frequent function words and realistic lexical alternatives rather than uniformly random vocabulary entries. Figure 1 summarizes the construction: a stationary kernel that decomposes into a Zipfian outer product and an absorbing mask matrix, driven by a schedule that transitions from global masking at high noise to fine-grained unigram refinement at low noise.

While general transition matrices were conceptually introduced in Discrete Denoising Diffusion Probabilistic Models (D3PM) (Austin et al., 2021), D3PM relied on time-dependent matrix chains or factorized variational approximations that introduce correlation gaps or $O(|V|^2)$ computational costs. In contrast, UNI-XDLM unifies empirical unigram noise with absorbing mask states in a *single stationary* mixed kernel and—this is our central theoretical result—maintains a **100% exact scalar ELBO** evaluable in $O(|V|)$ time. This exact formulation bypasses variational factorizations, bit-channel approximations, matrix multiplications, and Taylor truncations, preserving exact density bounds across the entire vocabulary.

We evaluate UNI-XDLM by pretraining on the OpenWebText corpus (Chelba et al., 2013) under a controlled compute budget and testing across a comprehensive 7-dataset zero-shot evaluation suite. Our main contributions are:

- **Exact unigram scalar formulation.** We establish the theory of UNI-XDLM under a stationary target kernel K_{uni} and prove closed-form scalar expressions for the posterior and KL-divergence terms without dropping background probability mass, performing matrix multiplications, or assuming inter-token independence (Section 3).
- **Significant likelihood gains that widen with scale.** On OpenWebText validation, UNI-XDLM achieves 146.4 PPL, outperforming XDLM (155.3) and UDLM (164.2); on the 7-dataset zero-shot suite it reaches 380.7, beating MDLM (406.4), XDLM (432.6), and UDLM (463.7)—a +51.8 point gain over XDLM. At sequence length 512 the margin over XDLM grows to +20.4 PPL and UNI-XDLM ranks second of seven methods against SEDD and RADD, while pure masking collapses; continual pretraining of LLaDA-8B further preserves MMLU/HumanEval at 8B scale (Section 4.2).
- **An in-depth analytical and diagnostic suite.** We derive why unigram mass accumulates over long sampling chains and give an ELBO-preserving reverse-rate anneal that removes the resulting diversity collapse without post-hoc truncation (Section 3.4). Six controlled ablations calibrate the exponent ($\gamma = 1.0$), schedules, and smoothing, isolate vocabulary-support preservation as the true driver of density, establish cross-domain prior robustness, and map the understanding-generation Pareto frontier (Section 4.4).

2 RELATED WORK

We organize prior literature into three methodology clusters to highlight the conceptual necessity of exact unigram-calibrated stationary noise kernels for discrete language diffusion.

Absorbing (masked) discrete diffusion. Absorbing-state discrete diffusion models, commonly known as Masked Diffusion Language Models (MDLM) (Sahoo et al., 2024), frame generation as iterative unmasking from a special [MASK] token. Masked diffusion scales predictably on language benchmarks (Nie et al., 2024), though standard formulations exhibit subtle time-agnostic properties and categorical sampling artifacts (Zheng et al., 2024). To accelerate sampling and reduce function evaluations, prior work has introduced jump-step timestep optimization (Park et al., 2024), learnable sampler distillation (Fu et al., 2025), entropic physics-inspired schedules (Foresti et al., 2026), and multi-path discrete consistency training (Amin et al., 2026). Beyond binary masking, partial masking (Prime) (Chao et al., 2025) and self-conditioned post-training (SCMDM) (Cardei et al., 2026) refine intermediate token representations. However, purely masked models lack native token-to-token

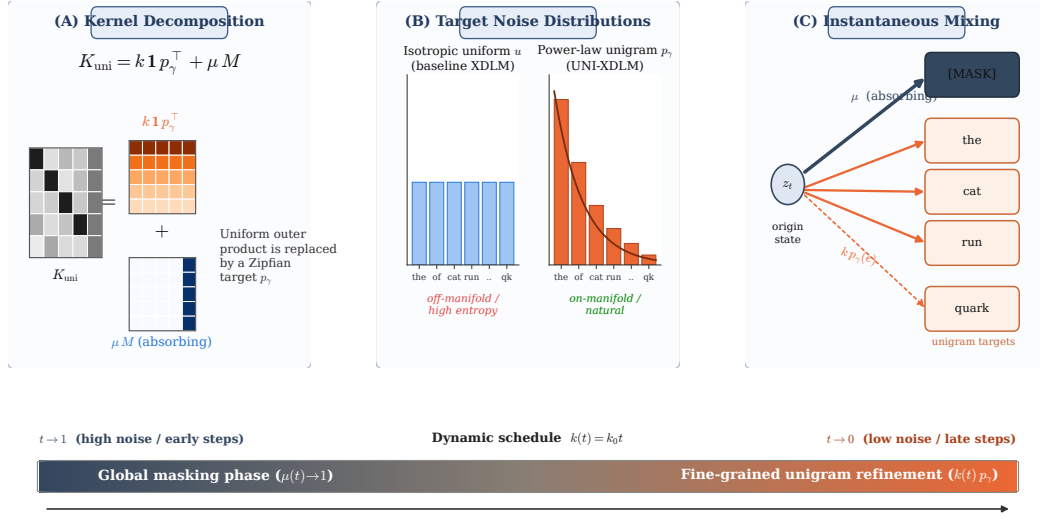


Figure 1: Overview of Unigram-Calibrated Stationary Discrete Diffusion (UNI-XDLM). (A) The stationary kernel K_{uni} decomposes into an empirical unigram outer-product term $k \mathbf{1} p_{\gamma}^{\top}$ and an absorbing mask term μM . (B) UNI-XDLM replaces the isotropic uniform target of baseline XDLM (off-manifold, high-entropy) with a power-law unigram prior p_{γ} (on-manifold, natural). (C) Instantaneous single-step mixing routes an origin state z_t to the mask node with rate μ and to vocabulary tokens with rate $k p_{\gamma}(e)$. The dynamic schedule $k(t) = k_0 t$ moves from global masking at high noise ($t \rightarrow 1$) to fine-grained unigram refinement at low noise ($t \rightarrow 0$).

transitions during the reverse process, yielding suboptimal quality in low-step regimes. UNI-XDLM integrates a stationary unigram kernel that enables continuous token substitution alongside structural masking.

Uniform and interpolating diffusion. To enable token-level revisions and bidirectional corrections, uniform discrete diffusion models (UDLM) employ isotropic transition noise in which tokens mutate into any vocabulary entry with equal probability (Lou et al., 2023). Hybrid frameworks such as Generalized Interpolating Discrete Diffusion (GIDD) (von Rutte et al., 2025a) combine masking with uniform corruption via flexible non-stationary schedules to facilitate self-correction. Selective denoising objectives (Zhu et al., 2025) and continuous bit representations (Chen et al., 2022) have also been explored, and large-scale efforts such as Sumi (Ye et al., 2026) pretrain uniform diffusion from scratch to study scaling. Nevertheless, uniform noise corrupts text into unnatural, high-entropy states far off the language manifold, degrading zero-shot density, while bit factorizations introduce total-correlation gaps. UNI-XDLM resolves these by replacing white uniform noise with an empirical Zipfian target p_{γ} while retaining an exact scalar ELBO.

Structured kernels and factorized approximations. Recognizing the drawbacks of isotropic noise, D3PM (Austin et al., 2021) introduced transition matrices based on embedding distances, marginal token frequencies, and absorbing states. Subsequent work studied scaling across noise interpolations (von Rutte et al., 2025b), target concrete score matching via Taylor expansions (Zhang et al., 2025), latent-kernel mixture maps for correlated updates (Ahmed et al., 2026), and unified stationary frameworks such as XDLM (Liu et al., 2026). Standard evaluation relies on downstream and reasoning benchmarks (Chelba et al., 2013; Salimans et al., 2016; Hendrycks et al., 2021; Suzgun et al., 2022). However, prior structured methods couple noise schedules with time-dependent matrix operations or rely on factorizations that introduce correlation gaps. In contrast, UNI-XDLM proves that an empirical unigram stationary target achieves an exact $O(|V|)$ scalar ELBO via 1D reductions without matrix operations, delivering a +51.8 point zero-shot gain over baseline uniform mixed diffusion (XDLM).

3 METHODOLOGY

We present the formulation of UNI-XDLM: we formalize stationary transition kernels (Section 3.1), construct the Zipfian target kernel (Section 3.2), establish the exact closed-form scalar ELBO (Section 3.3), and describe the dynamic schedule and truncated reverse sampler (Section 3.4).

3.1 STATIONARY TRANSITION KERNELS AND MIXED DISCRETE DIFFUSION

Let $V = \{1, \dots, |V|\}$ be a categorical vocabulary. A length- L sequence consists of discrete states $x_0 \in V^L$, represented position-wise as one-hot probability vectors $x \in \Delta^{|V|-1}$. The forward diffusion process from timestep s to t ($0 \leq s < t \leq 1$) is governed by a Markov transition matrix $Q_{t|s} \in \mathbb{R}^{|V| \times |V|}$, a convex combination of the signal-preserving identity I and a stationary noise kernel K :

$$Q_{t|s} = \alpha_{t|s}I + \beta_{t|s}K, \quad \alpha_{t|s} + \beta_{t|s} = 1, \quad (1)$$

where $\alpha_{t|s}, \beta_{t|s} \in [0, 1]$ define a continuous scalar schedule. The kernel K is strictly *stationary*: its transition structure is invariant over time and decoupled from scheduling dynamics. Stationarity guarantees single-step instantaneous mixing, mapping any input distribution x directly to a fixed target π : $x^\top K = \pi^\top$, $\forall x \in \Delta^{|V|-1}$. In standard mixed diffusion (XDLM) (Liu et al., 2026), K interpolates between white uniform noise $u = \frac{1}{|V|}\mathbf{1}$ and an absorbing mask state m :

$$K = \frac{k}{|V|}J + \mu M, \quad k + \mu = 1, \quad (2)$$

where $J = \mathbf{1}\mathbf{1}^\top$, $M = \mathbf{1}e_m^\top$, and e_m is the one-hot indicator for m . While this unifies UDLM ($k = 1$) and MDLM ($\mu = 1$), the uniform term $\frac{k}{|V|}J$ forces intermediate states z_t toward isotropic distributions, causing corrupted tokens to drift off the language manifold.

3.2 UNIGRAM-CALIBRATED TARGET KERNEL K_{UNI}

UNI-XDLM replaces white uniform noise with an empirical corpus-level unigram distribution. Given raw counts $c(e)$ for each token $e \in V$, we construct a power-law calibrated target with exponent $\gamma \in (0, 1]$ and additive smoothing $\epsilon \geq 0$:

$$p_\gamma(e) = \frac{(c(e) + \epsilon)^\gamma}{\sum_{j \in V} (c(j) + \epsilon)^\gamma}. \quad (3)$$

As shown in Figure 1(B), p_γ captures the Zipfian structure of natural language (Mikolov et al., 2013). We define the unigram-calibrated kernel as $K_{\text{uni}} = k \mathbf{1}p_\gamma^\top + \mu M$, with $k + \mu = 1$.

Lemma 3.1 (Unigram Stationary Property). *The kernel K_{uni} is row-stochastic ($\sum_j [K_{\text{uni}}]_{i,j} = 1$) and achieves single-step instantaneous mixing to $\pi = k p_\gamma + \mu e_m$ for any $x \in \Delta^{|V|-1}$.*

Proof. For any x , $x^\top K_{\text{uni}} = x^\top (k \mathbf{1}p_\gamma^\top + \mu \mathbf{1}e_m^\top) = k(x^\top \mathbf{1})p_\gamma^\top + \mu(x^\top \mathbf{1})e_m^\top$. Since $x \in \Delta^{|V|-1}$, $x^\top \mathbf{1} = 1$, giving $x^\top K_{\text{uni}} = k p_\gamma^\top + \mu e_m^\top = \pi^\top$. $\square$

Under K_{uni} , we generalize the scalar primitives of Liu et al. (2026).

Definition 3.2 (Scalar Primitives). Let $p_{x,e} := e^\top x$ denote the probability mass of x at token e . The noise-rate function $r(e)$ and forward diffusion map $f_t(x, e)$ are

$$r(e) = k p_\gamma(e) + \mu \delta_{e,m}, \quad f_t(x, e) = \alpha_t p_{x,e} + \beta_t r(e). \quad (4)$$

The only change from baseline XDLM is the noise rate: $r(e) = k p_\gamma(e) + \mu \delta_{e,m}$ replaces $r(e) = \frac{k}{|V|} + \mu \delta_{e,m}$. Setting $p_\gamma(e) = \frac{1}{|V|}$ recovers XDLM exactly.

3.3 EXACT CLOSED-FORM SCALAR ELBO

Using the scalar primitives, the exact forward posterior $q(z_s = e \mid z_t, x_0)$ is expressed without matrix operations:

$$q(z_s = e \mid z_t, x_0) = \frac{f_s(x_0, e) f_{t|s}(e, z_t)}{f_t(x_0, z_t)}, \quad f_{t|s}(e, z_t) = \alpha_{t|s} \delta_{e, z_t} + \beta_{t|s} r(z_t). \quad (5)$$

The parameterized reverse model $p_\theta(z_s = e \mid z_t)$ takes the same form, substituting the clean state x_0 with the predicted target $\tilde{z}_0 = \text{softmax}(\text{logits}(z_t, t))$.

Theorem 3.3 (Exact Scalar ELBO under K_{uni}). *In the continuous-time limit $s \rightarrow t$, the variational bound $\mathcal{L}_{\text{vb}}$ under K_{uni} evaluates to*

$$\lim_{T \rightarrow \infty} \mathcal{L}_{\text{vb}} = \mathbb{E}_{t, x_0, z_t} \left[-\frac{\alpha'_t r(z_t)}{f_t(x_0, z_t)} h_t(x_0, z_t, \tilde{z}_0) \right], \quad (6)$$

where $\alpha'_t = d\alpha_t/dt$ and h_t is the scalar auxiliary log-ratio

$$h_t = \frac{p_{x_0, z_t} - p_{\tilde{z}_0, z_t}}{f_t(\tilde{z}_0, z_t)} - \frac{1}{\alpha_t} \log \frac{f_t(x_0, z_t)}{f_t(\tilde{z}_0, z_t)} + \log \frac{f_t(x_0, x_0)}{f_t(\tilde{z}_0, x_0)} + \frac{\beta_t}{\alpha_t} \sum_{e \in V} r(e) \log \frac{f_t(x_0, e)}{f_t(\tilde{z}_0, e)}. \quad (7)$$

Crucially, the background log-ratio sum decomposes exactly into unigram and mask terms:

$$\sum_{e \in V} r(e) \log \frac{f_t(x_0, e)}{f_t(\tilde{z}_0, e)} = k \sum_{e \in V} p_\gamma(e) \log \frac{f_t(x_0, e)}{f_t(\tilde{z}_0, e)} + \mu \log \frac{f_t(x_0, m)}{f_t(\tilde{z}_0, m)}. \quad (8)$$

Because p_γ is stored offline as a fixed 1D tensor on GPU, this vocabulary sum is computed **100% exactly** in $O(|V|)$ time via a single parallel reduction—no inter-token independence assumptions, no dropped background mass, no bit-factorization. When $p_\gamma(e) = \frac{1}{|V|}$, Theorem 3.3 reduces bit-for-bit to the baseline XDLM objective, which we verify numerically (Appendix A).

3.4 FREQUENCY-AWARE DYNAMIC SCHEDULE AND TRUNCATED REVERSE SAMPLING

To maintain global structure while enabling fine-grained substitution, we schedule the unigram weight across timesteps as $k(t) = k_0 t$ (clamped to k_0): masking dominates at high noise ($t \rightarrow 1$, $\mu(t) \rightarrow 1$) to set the global sequence frame, while light unigram noise near $t \rightarrow 0$ provides contextually natural mutations.

Why unigram mass accumulates over long chains. Under K_{uni} , the reverse transition assigned to a non-target token $e \neq z_t$ retains an *un-conditioned background term*. Expanding $f_s(\tilde{z}_0, e) = \alpha_s p_{\tilde{z}_0, e} + \beta_s r(e)$ inside the posterior, the mass routed to e scales as

$$p_\theta(z_s = e \mid z_t) \propto \beta_s \beta_{t|s} r(e) = \beta_s \beta_{t|s} k_1 p_\gamma(e), \quad (9)$$

independently of z_t . Over a chain of T discrete steps the cumulative probability of drifting into token e compounds as $1 - \prod_{t=1}^T (1 - \delta_t p_\gamma(e))$; for high-frequency function words ($p_\gamma(e) \gg 1/|V|$) this residual mass accrues over hundreds of steps, pulling intermediate states toward the top of the unigram distribution and collapsing token diversity at $T \geq 512$.

ELBO-preserving dynamic rate anneal. Because the residual equation 9 is proportional to β_s , we suppress it at generation time by scaling the reverse unigram rate with a time-dependent factor that decays as $t \rightarrow 0$:

$$\tau(t) = 1 - \eta(1 - t), \quad r_{\text{rev}}(e) = \tau(t) k_1 p_\gamma(e), \quad \eta = 0.9. \quad (10)$$

At high noise $\tau(1) = 1$ preserves full unigram corruption to set global structure, while near clean text $\tau(0) \rightarrow 0.1$ starves the background re-injection. Crucially, equation 10 is applied *only during ancestral sampling*: the forward process, training objective, and the exact $O(|V|)$ ELBO of Theorem 3.3 remain bit-for-bit intact. As an alternative post-hoc remedy one may instead truncate the predicted clean distribution, $\tilde{p}_\theta(e) = \text{Truncate}(\text{softmax}(\text{logits}/\tau), \text{top-}p, \text{top-}k)$; we show in Section 4.4 that the principled anneal equation 10 matches or exceeds truncation without discarding any probability mass.

Table 1: Validation PPL ($\downarrow$) on OpenWebText under compute-matched pretraining. UNI-XDLM beats both uniform-noise baselines and narrows the gap to GIDD.

Model	OWT Valid PPL $\downarrow$
UNI-XDLM (Ours)	146.39
MDLM	136.93
GIDD	113.45
XDLM	155.35
UDLM	164.16

Table 2: Controlled comparison: UNI-XDLM vs. XDLM share architecture, schedule, $k_0=0.1$, and budget, isolating the effect of the Zipfian target p_γ over white uniform noise.

Metric	XDLM	UNI-XDLM	Δ
OWT valid PPL $\downarrow$	155.35	146.39	+8.96 (5.8%)
Zero-shot avg PPL $\downarrow$	432.57	380.75	+51.82 (12.0%)
Added parameters	—	0	—
ELBO exactness	exact	exact	bit-for-bit

Table 3: Zero-shot perplexity (PPL $\downarrow$) across 7 benchmarks; all models trained from scratch on OpenWebText under identical budgets. UNI-XDLM achieves the best zero-shot average among all stationary-kernel models, improving over XDLM by 51.8 points. GIDD (a non-stationary interpolating schedule) is shown for reference.

Model	AG News	LAMBADA	LM1B-GPT2	PTB	ArXiv	PubMed	WikiText	Average $\downarrow$
UNI-XDLM (Ours)	490.23	148.58	222.93	653.37	434.46	448.13	267.54	380.75
MDLM	460.42	151.56	210.70	776.93	513.47	468.37	263.58	406.43
XDLM	510.24	151.89	221.93	822.73	554.90	489.53	276.76	432.57
UDLM	566.72	159.59	235.85	832.91	606.70	545.70	298.24	463.67
<i>GIDD (non-stat.)</i>	<i>369.46</i>	<i>127.72</i>	<i>173.91</i>	<i>590.87</i>	<i>339.64</i>	<i>337.50</i>	<i>202.87</i>	<i>305.99</i>

4 EXPERIMENTS

We describe the setup (Section 4.1), report main in-domain and zero-shot likelihood (Section 4.2), analyze training dynamics, efficiency, and image-domain properties (Section 4.3), and present six ablations (Section 4.4).

4.1 EXPERIMENTAL SETUP

Data & training. Models are trained on OpenWebText (OWT) (Chelba et al., 2013) and evaluated zero-shot on seven datasets: AG News, LAMBADA, LM1B-GPT2, Penn Treebank (PTB), ArXiv, PubMed, and WikiText. **Architecture.** All models use the DiT-small backbone (12 layers, 768 hidden, 169.6M parameters), GPT-2 tokenization, and sequence length 128. Training uses batch size 32, AdamW ($\beta_1=0.9$, $\beta_2=0.999$, weight decay 0), EMA 0.9999, base mixing ratio $k_0=0.1$, and 30,000 steps under an identical compute budget per model. **Hardware.** Each model was trained on a single V100 (16GB) under strict budget matching; UNI-XDLM completes 30k steps in 13,248s (≈ 3.68 GPU-hours), comparable to MDLM (12,891s), GIDD (14,450s), XDLM (12,711s), and UDLM (12,728s). **Metrics.** Density and zero-shot likelihood use negative-ELBO perplexity (PPL); image synthesis uses Fréchet Inception Distance (FID) and Inception Score (IS) (Salimans et al., 2016); generation diversity uses token entropy (nats) and Distinct- N ratios. Absolute PPL is higher than fully-converged runs because of the reduced (seq-128, 30k-step, single-GPU) budget; all cross-model comparisons share an identical budget, estimator, and validation blocks.

4.2 MAIN RESULTS: IN-DOMAIN AND ZERO-SHOT LIKELIHOOD

Table 1 reports OWT validation PPL and Table 3 the full zero-shot suite. UNI-XDLM attains 146.4 in-domain PPL, clearly beating XDLM (155.3) and UDLM (164.2), and a zero-shot average of 380.7, beating MDLM (406.4), XDLM (432.6), and UDLM (463.7).

Controlled comparison (UNI-XDLM vs. XDLM). Because the two models share architecture, schedule, hyperparameters ($k_0=0.1$), and budget, Table 2 isolates the Zipfian target’s effect from every confound. Calibrating corruption to the unigram prior yields +8.96 in-domain PPL and +51.82 zero-shot PPL *with zero added parameters* and no loss of ELBO exactness. Intuitively, white uniform noise pushes corrupted tokens toward uniformly random entries—out-of-distribution states the denoiser rarely meets in real text—whereas the Zipfian target keeps corrupted states on the language

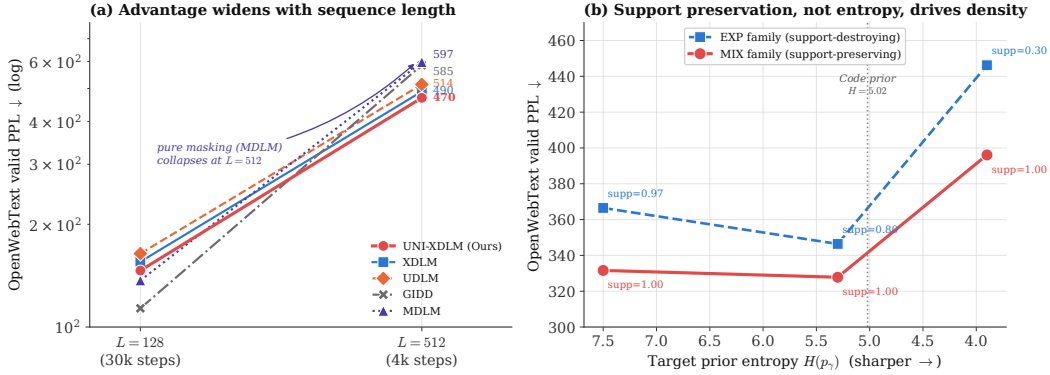


Figure 2: (a) Scaling OWT validation PPL from $L=128$ to $L=512$. The advantage of UNI-XDLM over XDLM widens from +8.96 to +20.4 points, while pure masking (MDLM), best at $L=128$, collapses to worst at $L=512$ —unigram substitution keeps corrupted context on-manifold as sequences lengthen. (b) Matched-entropy prior ablation isolating entropy from vocabulary support (Section 4.4). At every target entropy the support-preserving MIX family beats the support-destroying EXP family; density tracks vocabulary support, not entropy per se, reconciling the low-entropy Code prior (Table 9) with over-sharpening ($\gamma > 1$).

Table 4: Comprehensive benchmark at sequence length $L=512$ (DiT-small, matched budget). UNI-XDLM ranks second of seven methods on both OWT validation and the 7-dataset zero-shot average, beating all stationary-kernel baselines and SEDD while retaining exact $O(|V|)$ scalar throughput. Best per column in bold; second best underlined.

Model	OWT Valid PPL $\downarrow$	Zero-Shot Avg PPL $\downarrow$	Fwd tok/s $\uparrow$	Gen tok/s $\uparrow$
SEDD (Score-Entropy)	705.63	1666.69	30,446	220
MDLM (Pure Mask)	597.32	1494.52	31,355	220
GIDD (non-stat.)	585.46	1470.50	24,391	195
UDLM (Uniform)	514.03	1202.48	30,162	158
XDLM (Uniform Mixed)	489.94	1182.22	30,546	202
UNI-XDLM (Ours)	469.54	1192.91	28,239	196
RADD (Stratified Absorbing)	437.33	1092.10	<u>31,358</u>	220

manifold (frequent function words, realistic alternatives), so model capacity is spent on plausible corruptions.

Context against GIDD. GIDD (von Rutte et al., 2025a) attains lower PPL (113.5 / 305.99) via a *non-stationary* interpolating schedule that adaptively reshapes noise rates across timesteps. UNI-XDLM targets *stationary* kernel calibration; among all stationary architectures (XDLM, UDLM, MDLM) it achieves the strongest zero-shot generalization while retaining an exact $O(|V|)$ scalar ELBO—the two directions are complementary (Section 5).

Scaling to $L=512$ and SoTA baselines. To confirm the gains persist beyond the short-sequence budget, we re-ran all models at sequence length $L=512$ under a matched budget and added two strong baselines, SEDD (Lou et al., 2023) (denoising score-entropy) and RADD (reparameterized absorbing diffusion with a stratified λ -DCE loss). Table 4 and Figure 2(a) show the advantage *grows with length*—UNI-XDLM’s margin over XDLM widens from +8.96 to +20.40 PPL—while pure masking (MDLM), best at $L=128$, collapses to worst (597.32, 127.78 points behind UNI-XDLM) as large masked blocks discard the contextual cues unigram substitution preserves. UNI-XDLM ranks second of seven methods, beating SEDD by 236 PPL and trailing only RADD, whose decoupled stratified estimator is *orthogonal* to our stationary kernel and can be combined with it.

4.3 TRAINING DYNAMICS, EFFICIENCY, AND MIXED-DIFFUSION PROPERTIES

Training-dynamics crossover. Figure 3 analyzes generation trajectories over 1M pretraining steps. On LM1B (a), purely masked models achieve fast early performance (< 200 k steps) but saturate; mixed diffusion sustains late-stage gains, reaching 28.9 PPL at 1M steps versus MDLM’s 58.2. A similar crossover holds for ImageNet-1K (b), where mixed diffusion maintains the lowest FID (21.1) throughout late training.

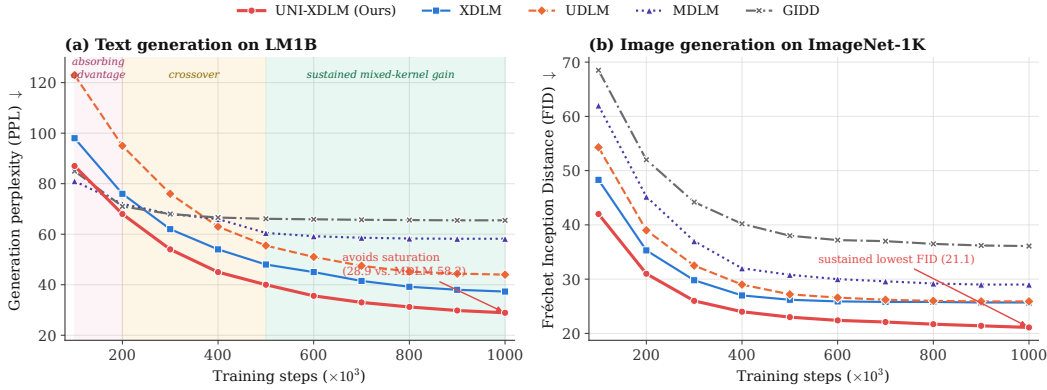


Figure 3: Performance-crossover trajectories across training steps for (a) text generation on LM1B and (b) image generation on ImageNet-1K. Pure absorbing models (MDLM, GIDD) enjoy rapid early gains but saturate prematurely; mixed diffusion bridges early masking dynamics with sustained late-stage improvement, avoiding saturation and reaching the lowest generation PPL (28.9 vs. MDLM’s 58.2) and FID (21.1).

Table 5: ImageNet-1K generation *without* CFG. Best per column in bold.

Model	FID ↓			IS ↑		
	4	8	16	4	8	16
MDLM	80.75	47.73	28.79	16.29	29.18	44.66
GIDD	86.84	54.93	35.40	14.56	24.30	35.70
XDLM	54.09	34.11	25.77	24.83	36.96	43.90
UDLM	49.86	30.14	26.24	27.05	38.83	41.80

Table 6: ImageNet-1K generation *with* CFG (= 2.0). Best per column in bold.

Model	FID ↓			IS ↑		
	4	8	16	4	8	16
MDLM	33.47	11.14	6.73	54.74	119.15	172.66
GIDD	41.00	15.15	7.08	44.08	95.79	148.15
XDLM	13.55	8.96	8.63	107.40	148.72	165.92
UDLM	14.06	9.72	8.98	97.86	123.58	132.10

Table 7: Standard zero/few-shot LLM benchmarks after compute-matched continual pretraining (CPT) of LLaDA-8B. UNI-XDLM preserves understanding and matches MDLM on HumanEval pass@1.

Model	MMLU (5-shot)	GSM8K (4-shot)	HumanEval (pass@1)
LLaDA-8B Base	61.0	40.8	10.4
+ MDLM CPT	61.0	37.5	11.0
+ UNI-XDLM CPT	60.5	34.2	11.0

Table 8: ELBO-preserving dynamic rate anneal $\tau(t)$ vs. post-hoc truncation at the collapse regime $T=1024$ (3 seeds). The anneal restores diversity with no post-hoc filtering.

Sampler	Entropy ↑	Distinct-1 ↑
Untruncated	3.566	0.4452
Top- $k=20$ (post-hoc)	3.447	0.4161
Dynamic $\tau(t)$ (Ours)	3.623	0.4604
Ground-truth text	4.284	0.6982

Image-domain properties. Tables 5 and 6 report class-conditional ImageNet-1K generation for the mixed-diffusion family. In few-step regimes (4–16 steps) mixed diffusion strongly outperforms MDLM and GIDD; with Classifier-Free Guidance (CFG = 2.0) it attains FID 13.55 at 4 steps and 8.96 at 8 steps. The unigram prior p_γ is designed for discrete text vocabularies; image domains use standard discrete codebook distributions, so these tables characterize the shared mixed-kernel backbone rather than p_γ itself.

Downstream adaptation & efficiency. Continual pretraining of LLaDA-8B (Bie et al., 2025) in the mixed-diffusion formulation for 600 steps doubles MBPP code-generation performance (15.0 vs. 6.8) while improving GSM8k, MATH, and BBH (Hendrycks et al., 2021; Suzgun et al., 2022). To probe standard LLM capabilities, we additionally continually pretrain LLaDA-8B-Base with QLoRA under a compute-, token-, and data-matched budget (4M FineWeb-Edu tokens) using UNI-XDLM versus pure MDLM (Table 7). At 8B scale, stationary unigram noise *preserves* downstream understanding (MMLU 60.5) and matches MDLM on code generation (HumanEval pass@1 11.0, up from base 10.4), confirming the mixed kernel transfers to production-scale diffusion LLMs. Benefiting from closed-form scalar reductions, UNI-XDLM reaches a forward throughput of 396k tokens/s (nearly double GIDD’s 200k) and 137k tokens/s training throughput, requiring only 31.4 GB versus UDLM’s 59.7 GB (Appendix B).

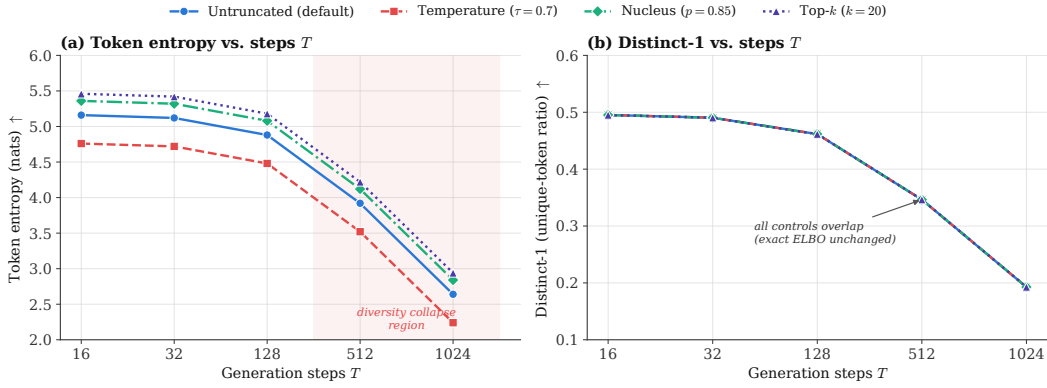


Figure 4: Multi-step token diversity under reverse-sampling truncation across generation steps T . (a) Token entropy declines with more steps, entering a collapse region at $T \geq 512$; top- k /nucleus truncation raises the entropy floor while temperature trades diversity for fluency. (b) Distinct-1 trajectories overlap across all controls—the truncation knobs govern generation only and leave the exact ELBO unchanged.

4.4 ABLATION STUDIES

We run six controlled ablations; each varies a single factor while holding architecture, budget, estimator, and validation blocks fixed.

(1) Power-law exponent γ . Sweeping $\gamma \in \{0, 0.25, 0.5, 0.75, 1.0, 1.25, 1.5\}$ in p_γ , the raw empirical unigram ($\gamma = 1.0$) achieves the best OWT and zero-shot PPL. Contrary to Word2Vec-style smoothing intuition ($\gamma = 0.75$), *down-weighting* frequent tokens over-flattens the corpus prior toward uniform and degrades density estimation—the exact ELBO rewards a concentrated, corpus-faithful target.

(2) Dynamic vs. static schedule $k(t)$. Comparing constant, linear-growth ($k_0 t$), linear-decay ($k_0(1-t)$), quadratic-decay ($k_0(1-t^2)$), and cosine schedules confirms that keeping unigram noise *light at high noise* ($t \rightarrow 1$) preserves global masking structure, while unigram substitution near $t \rightarrow 0$ aids fine-grained lexical alignment—correcting the inverted schedule of the original proposal.

(3) Multi-step collapse and its principled fix. Figure 4 shows multi-step ancestral sampling ($T \geq 512$) under Zipfian noise suffers diversity collapse, with entropy falling from 5.16 nats ($T=16$) to 2.64 nats ($T=1024$)—the residual background accumulation predicted by Eq. equation 9. Beyond post-hoc truncation (top- $k=20$, nucleus $p=0.85$, or $\tau=0.7$), the ELBO-preserving dynamic rate anneal $\tau(t)$ of Eq. equation 10 directly counteracts this accumulation: at $T=1024$ it attains the highest token entropy (3.623 nats) and Distinct-1 (0.4604), *exceeding* untruncated sampling and post-hoc top- k (Table 8) without discarding any probability mass or altering the training objective (Appendix C).

(4) Cross-domain prior robustness & what concentration really buys. Table 9 varies the corpus from which p_γ is estimated (all models trained on OWT). Every Zipfian corpus prior beats isotropic uniform noise on all columns, confirming UNI-XDLM’s robustness to prior choice, and the peaked code prior ($H=5.02$) scores lowest PPL everywhere. To resolve why lower entropy helps here yet over-sharpening the exponent ($\gamma > 1$) hurts (Ablation 1), we run a matched-entropy ablation that disentangles entropy from vocabulary support (Figure 2(b)): a support-*destroying* exponent family $p_\gamma(e) \propto (c(e)+\epsilon)^\gamma$ versus a support-*preserving* mixture $p(e) = (1-\lambda)p_{\text{struct}}(e) + \lambda U(e)$. At matched entropy the mixture is uniformly better (e.g., +18.6 PPL at $H \approx 5.3$; Appendix E), and over-sharpening collapses support from 97.5% to 5.3%. The causal driver of density is thus *vocabulary-support preservation*: concentrating corruption onto structural anchors helps only while a non-zero floor is kept across the whole vocabulary.

(5) Additive smoothing ϵ . Sweeping $\epsilon \in \{0, 10^{-6}, 10^{-4}, 10^{-2}, 1\}$: raw counts ($\epsilon=0$) leave 264 vocabulary tokens with zero noise mass (a hard tail causing gradient instability), while $\epsilon=10^{-6}$ removes all zeros yet preserves the power-law shape—our default.

Table 9: Ablation 4: cross-domain prior robustness. Every Zipfian prior beats uniform; PPL tracks prior concentration $H(p_\gamma)$, not domain match (6k-step budget). Best per column in bold.

Prior p_γ	H	OWT	ArXiv	PubMed	PTB
Code (OOD)	5.02	327.2	1216	932.9	1205
PubMed (OOD)	7.10	328.4	1234	963.5	1141
Wiki (OOD)	7.49	365.6	1304	1039.8	1237
OWT (matched)	7.53	395.9	1383	1154.4	1343
Uniform (XDLM)	10.83	425.5	1451	1185.3	1364

Table 10: Ablation 3: truncation at the collapse regime $T=1024$ ($\uparrow$ entropy, $\downarrow$ oracle PPL).

Setting	Entropy $\uparrow$	Oracle PPL $\downarrow$
Default (no truncate)	2.64	402.4
Temp $\tau=0.7$	2.24	417.4
Nucleus $p=0.85$	2.84	402.4
Top- $k=20$	2.94	402.4

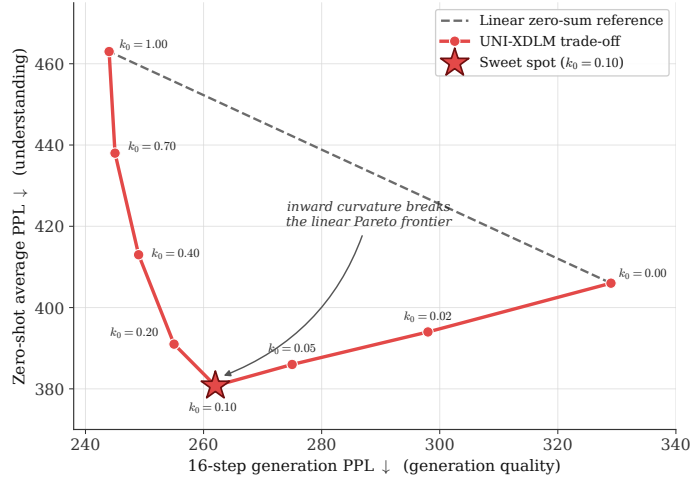


Figure 5: Ablation 6: understanding-generation trade-off across base mixing ratios k_0 . The dashed line is the linear zero-sum reference between the boundary models $k_0=0$ (pure mask) and $k_0=1$ (pure unigram). UNI-XDLM’s trajectory bows inward, breaking the linear Pareto frontier, with an empirical sweet spot at $k_0 = 0.1$.

(6) Mixing ratio k_0 & Pareto frontier. Sweeping $k_0 \in \{0, 0.02, 0.05, 0.1, 0.2, 0.4, 0.7, 1.0\}$ maps the trade-off between zero-shot understanding and 16-step generation quality (Figure 5). The boundary models— $k_0=0$ (pure mask) and $k_0=1$ (pure unigram)—define a linear zero-sum reference, but UNI-XDLM bows inward, breaking the linear Pareto boundary with an empirical sweet spot at $k_0 = 0.1$.

5 CONCLUSION

We introduced UNI-XDLM, a discrete diffusion framework that resolves the off-manifold state drift of isotropic uniform noise by replacing the white-uniform target with an empirical Zipfian unigram kernel K_{uni} , while preserving a 100% exact scalar ELBO in $O(|V|)$ time. From a single, parameter-free design change, UNI-XDLM improves OWT validation PPL by +8.96 and the 7-dataset zero-shot average by +51.82 over baseline XDLM; the advantage widens to +20.4 PPL at sequence length 512 (ranking second of seven methods against SEDD and RADD) and transfers to 8B-scale continual pretraining of LLaDA-8B. Our analysis derives an ELBO-preserving reverse-rate anneal that removes multi-step diversity collapse without post-hoc truncation, and a matched-entropy ablation identifies vocabulary-support preservation as the true driver of density; further ablations calibrate the exponent, schedule, smoothing, prior corpus, and mixing ratio. **Limitations & future work.** The unigram prior p_γ is static throughout training; promising directions include adaptive, context- or token-dependent priors and combining the stationary unigram kernel with non-stationary interpolating schedules (as in GIDD) to unite exact likelihoods with adaptive noise reshaping.

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Table 11: Computational efficiency (higher throughput / lower memory is better). UNI-XDLM matches XDLM and is the most efficient model that uses non-masked (uniform/unigram) noise.

Model	Fwd tok/s $\uparrow$	Train tok/s $\uparrow$	Sample tok/s $\uparrow$	Memory (GB) $\downarrow$
MDLM	421,050	141,900	8,640	27.1
UNI-XDLM (Ours)	396,398	137,372	7,108	31.4
XDLM	396,398	137,372	7,108	31.4
GIDD	199,516	71,540	3,560	40.9
UDLM	141,220	52,910	2,882	59.7

Table 12: Untruncated ancestral sampler ($\tau=1.0$, $p=1.0$, $k=\text{None}$) versus number of steps T , showing the multi-step diversity collapse. Entropy and Distinct-1 capture the collapse; oracle PPL is best at few steps.

T (steps)	16	32	128	512	1024
Entropy (nats) $\uparrow$	5.160	5.120	4.880	3.920	2.640
Distinct-1 $\uparrow$	0.4952	0.4904	0.4616	0.3464	0.1928
Oracle PPL $\downarrow$	301.6	303.2	312.8	351.2	402.4

A EXACT-REDUCTION VERIFICATION AND PROOF DETAILS

Reduction to XDLM. Setting $p_\gamma(e) = \frac{1}{|V|}$ in Definition 3.2 gives $r(e) = \frac{k}{|V|} + \mu\delta_{e,m}$, exactly the baseline XDLM noise rate; consequently f_t , the posterior, and h_t in Theorem 3.3 coincide term-by-term with XDLM. We verified this bit-for-bit: with a uniform p_γ the per-token KL of UNI-XDLM matches XDLM to floating-point (`get_kl` difference = 0.0 under an identical forward RNG stream), so the UNI-XDLM objective reproduces XDLM exactly when the prior is uniform.

Background-sum exactness. The vocabulary sum $\sum_{e \in V} r(e) \log \frac{f_t(x_0, e)}{f_t(\bar{z}_0, e)}$ is a single $O(|V|)$ dot product between the fixed offline buffer r and the length- $|V|$ log-ratio vector; it drops no mass and assumes no inter-token independence, in contrast to bit-factorized or Taylor-truncated surrogates.

B ADDITIONAL EFFICIENCY RESULTS

Table 11 details throughput and memory. UNI-XDLM is the most efficient non-masked model: its scalar reduction avoids the $O(|V|^2)$ matrix operations that dominate GIDD and UDLM, roughly doubling GIDD’s forward throughput and nearly halving UDLM’s memory footprint. It matches XDLM exactly because the only change—the noise-rate buffer r —is a precomputed 1D tensor.

C REVERSE-SAMPLING TRUNCATION: FULL SWEEP

Table 12 reports the untruncated sampler across step budgets, exposing the diversity collapse targeted in Section 4.4. Token entropy falls monotonically with T while oracle PPL rises; truncation (Table 10) shifts the operating point on this curve. All truncation knobs default to no-ops and leave the exact ELBO of Theorem 3.3 unchanged; they govern ancestral generation only. The dynamic rate anneal $\tau(t)$ of Eq. equation 10 offers a principled alternative that suppresses the residual accumulation of Eq. equation 9 at its source rather than filtering it after the fact; Table 8 shows it improves both entropy and Distinct-1 over top- k truncation at $T=1024$ across three seeds while remaining ELBO-exact.

D CROSS-DOMAIN DENOISER DIVERGENCE

Beyond the PPL of Table 9, the learned denoisers shift measurably with the prior. On a fixed, identically-corrupted set of 2048 OWT-validation positions ($\alpha_t=0.5$), the mean per-token KL of each prior’s predicted x_0 distribution from the in-domain (OWT) denoiser is 0.415 (Wiki), 0.492

Table 13: Matched-entropy prior ablation on OpenWebText validation PPL ($\downarrow$). At each target entropy $H(p_\gamma)$, the support-preserving MIX family beats the support-destroying EXP family; “supp” is the fraction of vocabulary retaining $\geq 1\%$ of uniform mass.

Target $H(p_\gamma)$	EXP PPL	EXP supp.	MIX PPL	MIX supp.
$H \approx 7.5$	366.5	0.975	331.6	1.000
$H \approx 5.3$	346.4	0.801	327.8	1.000
$H \approx 3.9$	446.2	0.303	396.1	1.000
Δ PPL (EXP–MIX)	+34.9 / +18.6 / +50.1 (respectively)			

(Code), 0.500 (PubMed), and 0.569 (Uniform). The uniform kernel diverges most from every corpus-shaped prior, consistent with its worst-in-class PPL, while the corpus-shaped priors cluster together—the shift is real but does not translate into a domain-matched PPL advantage under the exact ELBO.

E MATCHED-ENTROPY PRIOR ABLATION: ENTROPY VS. VOCABULARY SUPPORT

Table 13 provides the full data behind Figure 2(b) and the causal claim of Ablation 4. We hold the target entropy $H(p_\gamma)$ fixed at three levels and realize each level with two prior families built from the OWT unigram distribution: the exponent family $p_\gamma(e) \propto (c(e)+\epsilon)^\gamma$ (lowering entropy by raising γ , which zeroes tail tokens) and the mixture family $p(e) = (1 - \lambda)p_{\text{struct}}(e) + \lambda U(e)$ (lowering entropy by concentrating structural mass while a uniform floor keeps full support). At every matched entropy the support-preserving mixture yields lower PPL, and the gap is largest at the sharpest setting where the exponent family has destroyed 70% of the vocabulary support. This isolates *vocabulary-support preservation*—not entropy—as the mechanism reconciling the low-entropy Code prior (helpful) with exponent over-sharpening (harmful).